



FASR Science Case # WG3-1

Investigation of Compact Transient Structures in the Low Corona

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1 Science Goal(s)

The quiet Sun and coronal holes contain weak reconnection events that may contribute to coronal heating and solar-wind acceleration, but their accelerated electrons are rarely observed directly. The science test is whether spatially resolved microwave spectra and circular polarization can distinguish gyrosynchrotron emission from thermal free-free emission in compact transients, and whether the resulting occurrence and energy distributions are consistent with a significant contribution to coronal heating.

2 Scientific Rationale

2.1 Scientific Importance

The quiet Sun corona and coronal holes host a variety of phenomena that operate over a range of spatial and temporal scales. On the longer timescales, we have structures like coronal bright points, which can persist over days and are believed to be miniature active regions. A large variety of transients like mini-jets, mini-filament eruptions, etc. are also observed at smaller timescales. While there has been a lot of exploration on the thermal properties of these phenomena, there have been very limited investigation into their nonthermal properties. Understanding both the thermal and nonthermal properties of these transients are important for answering key questions like coronal heating and solar wind acceleration. This will also allow investigation of particle acceleration and magnetic reconnection theories at very different regimes than generally done.

2.2 Uniqueness to FASR Capabilities

Is this science case uniquely addressed by FASR? Why can't other facilities address this science and achieve the same goal?

Depending on frequency, the brightness temperatures of these transients range from about 0.003 MK to several MK above the quiet Sun value (Fu et al. 1987; Habbal et al. 1986; Mondal et al. 2023; see the appendix). Multiple transients may be present on the disk at once. Earlier observations were limited by deconvolution errors and sparse frequency sampling, so they favored the brightest

events and could not determine their spectra. The requirement is therefore set by completeness and classification: recover sources down to the simulated brightness levels, locate them across the disk, and obtain enough spectrum and Stokes V to separate nonthermal from thermal emission. The array design must be tested with injected sources on realistic quiet-Sun backgrounds; a claimed dynamic range is useful only if that test recovers source counts and spectra without a systematic bias.

2.3 Synergies

Describe potential synergies/complementarities between this FASR science case and those from current/future/planned facilities at all wavelengths (e.g., DKIST, MUSE, FIERCE, COSMO, ngGONG, etc.).

The lower corona is observed across wavelengths including the EUV, UV and the X-ray wavelengths. To do a detailed investigation into the energetics of these events and also to be a spectral modelling of the emissions, we need to obtain data from other wavebands. Thermal energetics obtained from EUV will be essential to understand the energy partition of these events. In the past, spectral line observations have been explained based on beam heating of nonthermal electrons (Testa et al. 2014). With the availability of FASR, we would be able to directly detect the nonthermal electron beams, which can then be put into models and directly compared with the UV observations.

2.4 Measurements Required by FASR

Provide a description of the necessary measurements to be carried out by the FASR to adequately address this science case. Please coordinate these measurements with the Science Requirements table in Section III.

The baseline product is full-disk Stokes I and V imaging every 10 s. It must detect the author-simulated excess brightness of 0.003 MK near 8 GHz and 0.05 MK near 1 GHz, with spectral sampling fine enough to distinguish the modeled gyrosynchrotron and free-free slopes. A 1-minute product is sufficient for occurrence statistics, but the shorter cadence is needed to associate emission with individual transients.

III. Science Requirements Tables

(A) Observational Target Description

Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.

One of the regularly observed transients observed at GHz frequencies are the coronal bright points. All of the above description is based on those earlier observations (Fu et al. 1987, Habbal et al. 1986, Mondal et al. 2023, etc.).

(A) OBSERVATIONAL TARGET		
Type of observation (what defines a ‘target’)	Provide a brief description of the target. Gyrosynchrotron emission due to nonthermal particles accelerated from small scale magnetic reconnections in the low solar corona.	
Number of targets	20–30	
Size of a single target (arcsec x arcsec)	20''–30'' at 1 GHz, 5''–15'' at 3 GHz	
Distribution of all targets (arcmin x arcmin)	Throughout the solar disc	
Peak brightness (sfu/beam or Kelvin)	N/A	
RMS brightness (sfu/beam or Kelvin)	0.001 MK at frequency ~ 8 GHz, 0.01 MK at ~ 1 GHz	
Expected circularly polarized flux density	Stokes V (sfu/beam)	
	V/I	20%
Expected linearly polarized flux density	Stokes Q or U (sfu/beam)	NA
	Q/I or U/I	NA

(B) Spectro-Temporal Requirements Discussion

Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.

The 0.8–8 GHz range follows the source models and the published observations cited below: it spans the frequencies where the modeled gyrosynchrotron excess is separable from the thermal background. The channelization must resolve the modeled spectral turnover and the cadence must preserve the event duration; these are science requirements, while the correlator implementation is a design choice.

(B) SPECTRAL-TEMPORAL REQUIREMENTS	
Central Frequency (GHz)	0.8–8
Instantaneous Bandwidth (GHz/pol)	The sequence must cover the modeled turnover and both spectral slopes within 10 s; backend allocation is a design choice.
Spectral resolution [MHz]	About 2% fractional width; confirm the value with source-injection classification tests.

(B) SPECTRAL-TEMPORAL REQUIREMENTS (continued)	
Temporal resolution (in seconds)	10

(C) Polarization Product Discussion

Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.

Both Stokes I and V are required. Since only about 10–15% Stokes V accuracy is desired, Stokes Q and U data products are not essential.

(C) POLARIZATION DATA PRODUCTS REQUIRED	
y	Stokes I
n	Stokes Q
n	Stokes U
y	Stokes V

(D) Imaging Requirements Discussion

Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.

The detection test is recovery of a compact excess above a structured quiet-Sun background, followed by spectral and polarization classification. The cited simulations give the brightness levels; they do not justify filtering all structure above a chosen baseline length as an architectural requirement. A design-phase injection/recovery study should determine how background modeling, imaging fidelity, and source crowding affect completeness. The 10 percent polarization accuracy is sufficient for the first test of whether a source is polarized; quantitative polarization inference may require better accuracy for the weakest events.

(D) IMAGING REQUIREMENTS	
Required angular resolution (arcsec) (single value or range)	About 30'' at 1 GHz and 5''–15'' above 3 GHz, based on the observed source sizes.
Largest angular scale required (arcsec)	At least 100'' for local background estimation around compact sources; source recovery must be verified without assuming that larger quiet-Sun structure can be discarded.
Mapped image size (arcmin x arcmin)	32 × 32
Required pixel resolution (arcsec)	1''
Number of output/image channels	Set by the source-model spectral sampling; approximately 100–150 samples over 0.8–8 GHz.
Output bandwidth (minimum and maximum frequency - GHz)	0.8–8

(D) IMAGING REQUIREMENTS (continued)		
Channel width (MHz)	About 2% fractional width; use finer sampling if source-injection tests show classification bias.	
Required rms (sfu/beam or Kelvin) [per channel] (if polarization products required define for each)	0.01 MK in Stokes I. 10% of Stokes I rms is required for Stokes V.	
Dynamic range within image (if polarization products required define for each)	At least 1000:1 in Stokes I for the modeled MK-level background and 0.003 MK excess; verify with injected sources.	
Polarization accuracy (%)	10%	
Zero spacing/total power required?	Yes. Provide calibrated total-power/zero-spacing information across the full 0.8–8 GHz science band, synchronized with the 10 s Stokes I/V products. This supports absolute fluxes and diffuse spectra for gyrosynchrotron and free-free emission; combine it with interferometric data to recover source structure. At 2.8 GHz it also provides the F10.7 reference.	
Required maximum latency (in seconds, or N/A)	N/A	
Required flux density scale calibration accuracy		1-3%
		5%
	x	10%
		20-50%

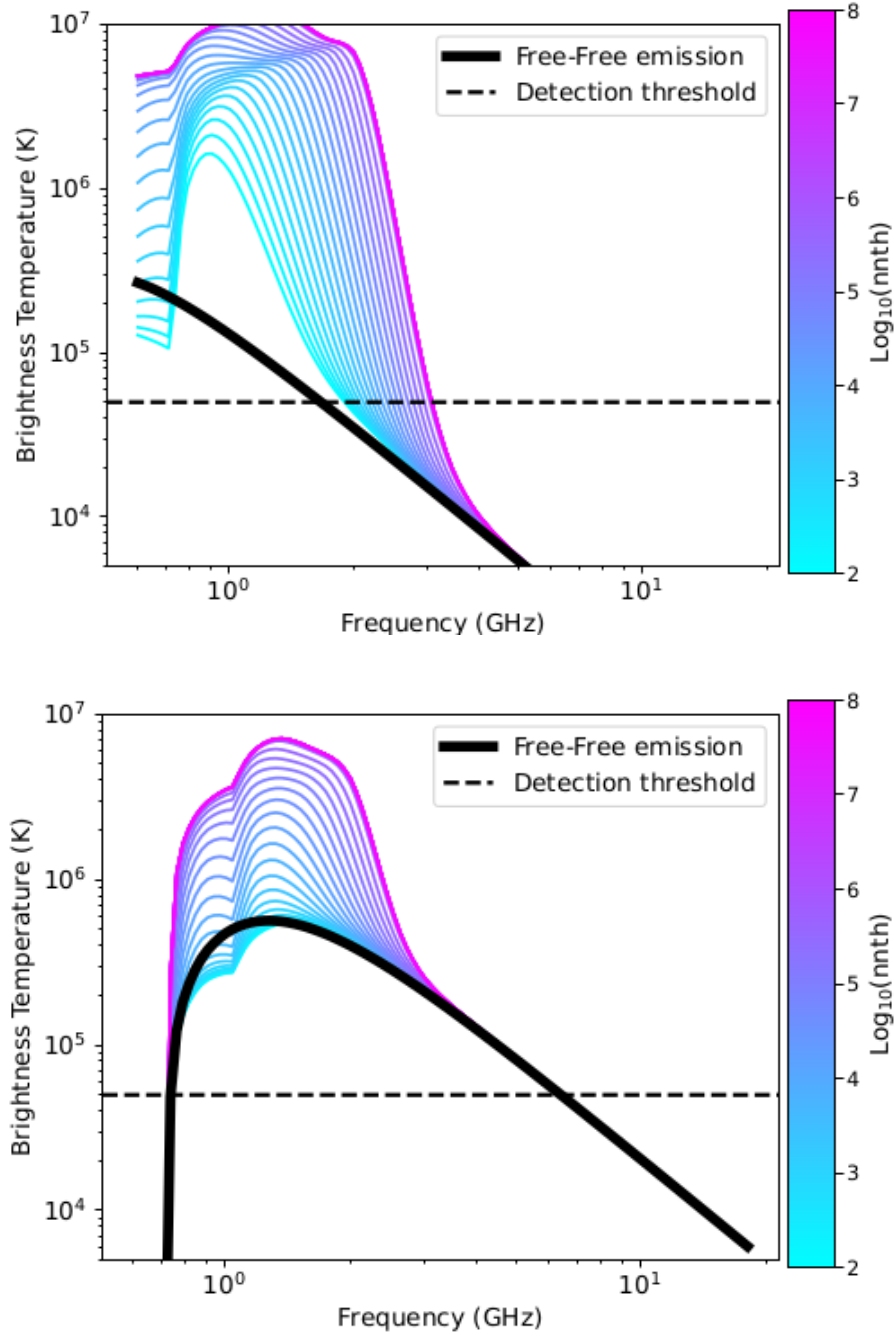
2.5 Other Performance or Functional Requirements

3 Appendix

Please provide any other relevant material necessary to understand and substantiate this Science Case.

Image Sensitivity Calculations

Shows the GS emission due to nonthermal electrons (shown with color) to that of thermal emission (shown in black). The distribution of nonthermal electrons follows a powerlaw with the minimum energy being 6 keV and the powerlaw index of 6. The LOS depth is 3 Mm. Temperature is set to 1 MK. Left and middle panel, the thermal density is set to $10^{9.2} \text{ cm}^{-3}$. In right panel, the thermal density is set to $10^{9.8} \text{ cm}^{-3}$. For the left panel, magnetic field is set to 100 G, whereas for the middle and right panels, the magnetic field is set to 200 G.



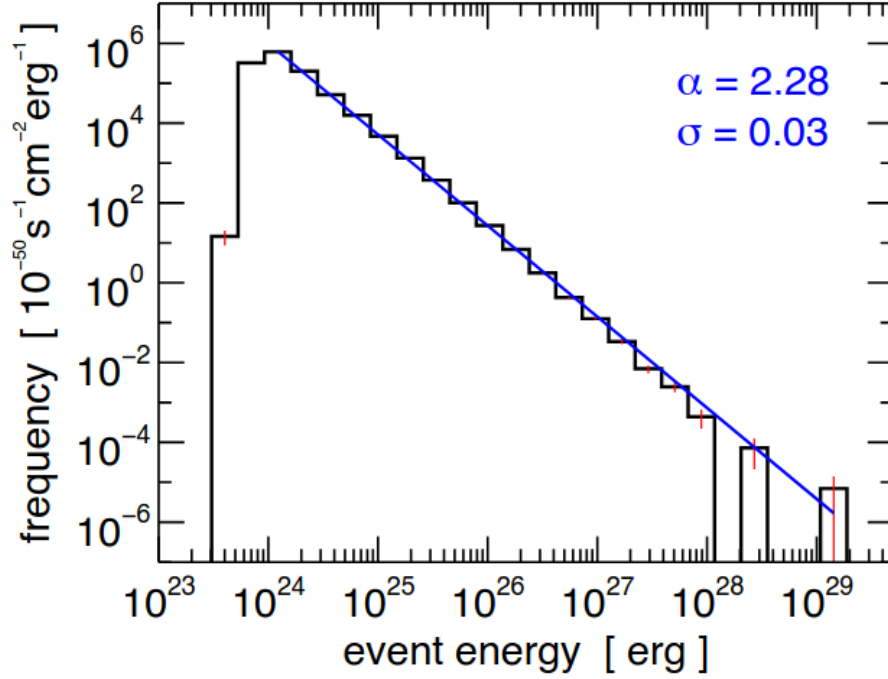
In the figure above, we have simulated the predicted brightness temperatures for different typical GS parameters. Magnetic field above coronal bright points typically range from 50–200 G. The nonthermal electron parameters chosen correspond to values estimated in [Mondal et al. 2023](#), [Glesener et al. 2020](#). It is evident that for low thermal density, we can separate out the GS emission from the thermal emission for lower magnetic field. With increase in thermal density, the required magnetic field also increases. The black dashed line shows 0.05 MK, which is the detection threshold which is being targeted.

Image Resolution

Coronal bright points typically have an angular scale of $20''$ – $30''$ at 1 GHz, so the completeness test should recover sources at about $30''$ resolution there (Habbal et al. 1986). At higher frequencies, transient sources as small as $5''$ have been reported. The relevant test is recovery of those source sizes and spectra on a structured quiet-Sun background, including cases in which the background varies across the source. Background estimation may be part of the analysis, but its success must be measured by source completeness and thermal/nonthermal classification rather than assumed as an imaging capability.

Estimating Number of Transients in a Single Image

Glesener et al. (2020) determined the nonthermal and thermal energy associated with a microflare, using HXR data. It is probably the only measurement available from HXR. The total nonthermal energy deposited was estimated to be 4×10^{29} ergs, whereas the corresponding associated thermal energy as measured by AIA was about 4×10^{28} ergs. Mondal et al. (2020) found that the ratio of the nonthermal and thermal energy associated with a weak radio transient was approximately 1. Thus based on the available literature, we estimate the ratio between non-thermal and thermal energies can vary between 1–10. Using a ratio of 1, the flare frequency distribution from Purkhart & Veronig (2022), angular resolution of $15''$, and duration of 10 s, we estimate that there are about 2000 transient event above nonthermal energy of 10^{24} ergs, over the solar disc. Similarly we estimate that there are about 200 events above an energy of 10^{25} ergs every 10 s. To relate these number numbers to the expected radio flux, we aim to recover, we use a set of typical coronal parameters. Using a magnetic field of 100 G, thermal electron density of $10^{9.2} \text{ cm}^{-3}$, nonthermal electron density of 10^6 , $e_{\min}$ of 5 keV, $e_{\max}$ of 10 MeV, nonthermal electron powerlaw index of 7, we find that the expected brightness temperature is approximately 0.05 MK. Using a PSF size of $10'' \times 10''$, we estimate that the total nonthermal energy dumped over a time interval of 10 s is 10^{26} ergs. This implies that it is expected there will be about 20–30 events (using the Purkhart & Veronig 2022 scaling relation) every 10 s with the total nonthermal energy dumped by those responsible electrons being above 10^{26} ergs.



Reference

- Fu, Q., Kundu, M. R., & Schmahl, E. J. 1987. Coronal bright points at 6 cm wavelength. *Solar Physics*, 108, 99–111.
- Habbal, S. R., Ronan, R. S., Withbroe, G. L., Shevgaonkar, R. K., & Kundu, M. R. 1986. Solar coronal bright points observed with the VLA. *The Astrophysical Journal*, 306, 740–750.
- Mondal, S., Oberoi, D., Biswas, A., & Kansabanik, D. 2023. Characterizing the spectral structure of weak impulsive narrowband quiet Sun emissions. *The Astrophysical Journal*, 953, 4.
- Testa, P., De Pontieu, B., Allred, J., et al. 2014. Evidence of nonthermal particles in coronal loops heated impulsively by nanoflares. *Science*, 346, 1255724.
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- Purkhart, S., & Veronig, A. M. 2022. Nanoflare distributions over solar cycle 24 based on SDO/AIA differential emission measure observations. *Astronomy & Astrophysics*, 661, A149.